

Empiricist

WEIGHT 1 · Kanta seat · LONGEST

Reproduce scripts · BOM · latency recipe · ablation ranks 1–8 · RC0–RC4 · GPU honesty

Claim under measurement

Can GoPro+printed gripper + IMU-SLAM + latency-matched relative-trajectory Diffusion Policy transfer dynamic/bimanual/long-horizon human demos to UR5/Franka with reproducible directional rankings?

Success bar = directional ranking — NOT paper 100%/87.5% parity

Shared spine + tattoo strip

Paste into every deck; Empiricist owns the numbers

- 1 In-the-wild teaching without in-the-wild robots
- 2 Relative SE(3) + latency matching + wrist fisheye → zero-shot UR5/Franka
- 3 Complementary to DROID (robot-in-wild) and ALOHA (lab teleop)
- 4 $\pi_{0.5}$ / GR00T CLEAR-cite UMI
- 5 Motion: $T_a=6$ @10Hz DP chunks; relative not abs/delta; latency first-class for dynamic

Reproduce scripts table

Official MIT repo — [CLEAR from GitHub](#) [README](#)

Main repo	github.com/real-stanford/universal_manipulation_interface
Env	mamba env create -f conda_environment.yaml → umi (Ubuntu 22.04)
SLAM	python run_slam_pipeline.py <session_dir> (+ Docker chicheng/orb_slam3)
Dataset	scripts_slam_pipeline/07_generate_replay_buffer.py → dataset.zarr.zip
Train 1×GPU	train.py --config-name=train_diffusion_unet_timm_umi_workspace (3090 24GB)
Train multi	accelerate --num_processes <n> train.py --config-name=...
Wild data	wget ../cup_in_the_wild.zarr.zip
Wild ckpt	wget ../cup_wild_vit_l_1img.ckpt
Eval	eval_real.py --robot_config=... -i <ckpt> -o <outdir> (C start / S stop)
Deploy forks	umi-on-legs · umi-arx (ARX X5)

Hardware BOM + deploy twin

73printed + 298 GoPro · 780 g · WSG-50 twin

\$73

3D-printed gripper BoM

\$298

GoPro Hero9 + accessories

780 g

mass · L310×W175×H210

155°

fisheye · raw (no undistort)

WSG-50

Schunk twin on robot

95A TPU

soft rib fingers · series-elastic

80 mm

finger stroke

UR5 / FR2

validated arms · same ckpt

Stream: Media Mod HDMI → Elgato HD60X UVC · IMU via GPMF · continuous width via ArUco fiducials

Fig. 2 — hardware / twin geometry

Handheld GoPro gripper ↔ robot wrist twin (observation alignment)

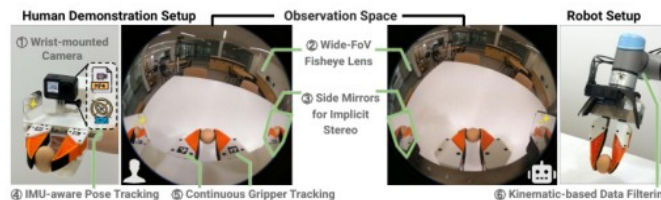


Fig. 2: UMI Demonstration Interface Design. Left: Hand-held grippers for data collection, with a GoPro as the only sensor and recording device. Middle: Image from the GoPro's 155° Fisheye view. Note the physical side mirrors highlighted in green which provide implicit stereo information. Right: UMI-compatible robot gripper and camera setup make observation similar to hand-held gripper view.

precise global action due to scale ambiguity, motion blur, or insufficient texture, which significantly restrict the precision of tasks for which the system can be employed.

- **Latency discrepancies:** During hand-held data collection, observation and action recording occur without latency. However, during inference, various latency sources, including sensor, inference, and execution latencies, arise within the system. Policies unaware of these latency discrepancies will encounter out-of-distribution input and in turn, generate out-of-sync actions. This issue is especially salient for fast and dynamic actions.

- **Insufficient policy representation:** Prior works often use simple policy representations (e.g., MLPs) with action regression loss, limiting their capacity to capture complex multimodal action distributions inherent in human data. Consequently, even with precisely recovered demonstrated actions and all discrepancies removed, the resulting policy could still struggle to fit the data accurately. This further hampers large-scale, distributed human data collection, as more demonstrators increase action multimodality.

In this paper, we address these issues with careful design of the demonstration and policy interface:

- First, we aim to identify the right **physical interface** for human demonstration that is intuitive and meanwhile able to capture all the information necessary for policy learning. Specifically, we use a Fisheye lens to increase the field of view and visual context, and add side mirrors on the gripper to provide implicit stereo observation. When combined with the GoPro's built-in IMU sensor, we can enable robust tracking under fast motion.
- Second, we explore the right **policy interface** (i.e., observation and action representations) that could make the policy hardware-agnostic and thereby enable effective skill transfer. Concretely, we employ inference-time latency matching to handle different sensor observation and execution latency, use relative trajectory as action representation to remove the need for precise global action, and finally, apply Diffusion Policy [9] to model

multimodal action distributions.

The final system, **Universal Manipulation Interface (UMI)**, provides a practical and accessible framework to unlock new robot manipulation skills, allowing us to demonstrate any actions in any environment while maintaining high transferability from human demonstration to robot policy.

With just a wrist-mounted camera on the hand-held gripper (Fig. 2), we show that UMI is capable of achieving a wide range of manipulation tasks that involve dynamic, bimanual, precise and long-horizon actions by only changing the training data for each task (Fig. 1). Furthermore, when trained with diverse human demonstrations, the final policy exhibits zero-shot generalization to novel environments and objects, achieving a remarkable 70% success rate in out-of-distribution tests, a level of generalizability seldomly observed in other behavior cloning frameworks. We open-source the hardware and software system at <https://umi-gripper.github.io>.

II. RELATED WORKS

A key enabler for any data-driven robotics system is the data itself. Here, we review a few typical data collection workflows in the context of robotic manipulation.

A. Teleoperated Robot Data

Imitation learning learns policies from expert demonstrations. Behavior cloning (BC), utilizing teleoperated robot demonstrations, stands out for its direct transferability. However, teleoperating real robots for data collection poses significant challenges. Previous approaches utilized interfaces such as 3D spacemouse [9, 54], VR or AR controllers [15, 3, 13, 19, 31, 51, 12], smartphones [44, 43, 22], and haptic devices [38, 47, 43, 26, 4] for teleoperation. These methods are either very expensive or hard to use due to high latency and lack of user intuitiveness. While recent advancements in leader-follower (i.e. puppeting) devices such as ALOHA [53, 15] and GELLO [46] offer promise with intuitive and low-cost interfaces, their reliance on real robots during data collection limits the type and number of environments the system can gain access to for “in-the-wild” data acquisition. Exoskeletons [14, 20] remove the dependence on real robots during data

Paper p.2 · 155° fisheye + side mirrors + matching cam-finger geometry

Ablation shortlist ranks 1-8

AUTHORITATIVE: [umi-2402.10329-ablations.md](#)

#1	Rel / Abs / Delta	100% / 25% / 80%	Cup · PRIMARY smoke
#2	Latency on/off	87.5% → 57.5%	Toss · PRIMARY smoke
#3	Wild vs narrow	71.7% vs 0%	1400/30 · Partial
#4	Fisheye vs 69°	100% vs 55%	Partial FoV crop
#5	Mirrors n/raw/refl	90 / 85 / 100%	N (needs crop)
#6	ViT vs ResNet	70% vs 0%	Dish · defer full
#7	Inter-grip proprio	70% → 30%	Cloth · Partial
#8	UR5 → Franka	100% → 90%	Same ckpt · stub

RC1 — relative action (#1 PRIMARY)

Cup §V-A · Rel 20/20 · Delta 16/20 · Abs 5/20 · 305 demos

Relative traj

100%

SE(3) w.r.t. EE @ to
whole chunk · PD2.1

Delta

80%

step-to-previous
error accumulates

Absolute

25%

world/base frame
calib bias brittle

Pass (course): Rel best under origin-jitter OOD; Rel $\geq$ Delta on clean ID; Abs shows calib-bias —
DIRECTIONAL, not 100/80/25 real SR

Fig. 6 — relative / delta / absolute

Action representation definitions (paper)

For each observation stream, we individually measure their latency (details see §A1-A3). At inference time, we align all observations with respect to the stream with the highest latency (usually the camera). Specifically, we first temporally down-sample the RGB camera observations to the desired frequency (often 10-20Hz), and then use the capture timestamp of each image t_{obs} to linearly interpolate gripper and robot proprioception streams. In bimanual systems, we soft-synchronize two cameras by finding the nearest neighbor frames, which can be off by a maximum of $\frac{1}{60}$ seconds. The result is a sequence of synchronized observations that conform to UMI policy, shown in Fig. 5 (a).

PD1.2) Action latency matching. UMI policy assumes the output as a sequence of synchronized EE poses and gripper widths. However, in practice, robot arms and grippers can only track the desired pose sequence up to an **execution latency**, that varies across different robot hardware. To make sure the robots and grippers reach the desired pose at the desired time (given by the policy), we need to send commands **ahead of time** to compensate for execution latency, as shown in Fig. 5 (c). See §A4 for execution latency calibration details.

During execution, the UMI policy predicts the action sequence starting at the last step of observation t_{obs} . The first few actions predicted are immediately outdated due to observation latency $t_{obs} - t_{obs}$, policy inference latency $t_{out} - t_{obs}$ and execution latency $t_{act} - t_{out}$. We simply discard the outdated actions and only execute actions with the desired timestamp after t_{act} for each hardware.

PD2. Relative end-effector pose. End-effector (EE) pose is central to both UMI's observation and action space. To avoid dependence on embodiment/deployment-specific coordinates, we represent all EE poses relative to gripper's current EE pose.

PD2.1) Relative EE trajectory as action representation.

Prior works have shown the significant impact of action space selection on task performance [9], with experimental evidence favoring absolute positional actions over delta actions. However, we found that a relative trajectory representation, defined for an action sequence starting at t_0 as a sequence of $SE(3)$ transforms denoting the desired pose at t relative to the initial EE pose at t_0 , allows the system to be more robust against tracking errors during data collection and camera displacements.

PD2.2) Relative EE trajectory as proprioception. Similarly, we represent the proprioception of history EE poses as a relative trajectory. When observation horizon is set to 2, this representation effectively provides velocity information to the policy. Combined with our wrist-mounted camera observation space, relative trajectory allows our system to be **calibration-free**. Moving the robot base during execution will not affect task performance (Fig. 10 (a)), as long as the objects are still within reach range, making the UMI framework applicable to mobile manipulators as well.

PD2.3) Relative inter-gripper proprioception. When using UMI in a bimanual setup, we found that providing the policy with the relative pose between the two grippers to be critical for bimanual coordination and task success, as shown

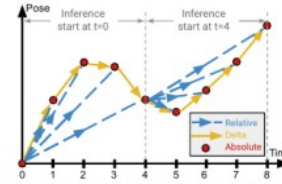


Fig. 6: Relative Trajectory as Action Representation. Relative trajectory, used by UMI, is a sequence of end-effector (EE) poses relative to the **same current EE pose** for each inference step. In contrast, **Delta actions** represents each action step relative to its immediate previous action, therefore accumulates error. **Absolute action** requires a global coordinate frame for all actions, which is difficult to define for in-the-wild data collection.

in Sec. V-C. The effect of inter-gripper proprioception is particularly large when the visual overlap between two cameras is small. The inter-gripper proprioception is enabled by our map-then-localize data collection scheme that constructs a scene-level global coordinate system HD4. For each new scene, we first collect a video that builds a map for the scene. Then, all demonstrations collected in this scene are relocalized to the same map, therefore sharing the same coordinate system. Despite the videos from each gripper being relocalized separately, the relative pose between two grippers at each time step can be calculated using their shared coordinates.

IV. EVALUATIONS

In our experiment, we aim to evaluate the UMI framework's effectiveness for deployable policy learning in three aspects:

- **Capability:** How well can we transfer UMI demonstrations to effective robot policy? Especially for complex, dynamic, bimanual, and long-horizon manipulation skills.
- **Generalization:** Will data collected in the wild within diverse environments help the policy to generalize to unseen environments and objects?
- **Data collection efficiency:** How fast can we collect manipulation data with UMI? What's the accuracy of the SLAM system?

To access capability and generalization, we evaluate UMI on 4 real-world robotic tasks across both narrow domain and in-the-wild environments, shown in Fig. 7. To measure data collection efficiency, we compare the UMI gripper with human hand demonstration and a typical teleop interface. See §B for detailed data collection protocol.

V. CAPABILITY EXPERIMENTS

We study UMI's ability to capture and transfer single-hand, bimanual, dynamic, and long-horizon manipulation skills with four tasks. For capability experiments, all tasks are evaluated in the same environment as data collection but with randomized robot and object initial states. To ensure a fair comparison,

Relative trajectory preferred — not delta-to-previous, not world-absolute

RC2 — latency matching (#2 PRIMARY)

Toss §V-B · same ckpt · $105/120 = 87.5\%$ vs $69/120 = 57.5\%$

87.5%

Ours · latency matched

57.5%

No matching · -30 pp

20 Hz

toss · 1.0× speed · ResNet-34

- Same trained checkpoint — inference-time flag only (set measured latencies $\rightarrow 0$)
- Grid: {zero, measured_match} + extended {0.5×, 2×, sign_flip} + obs delay $d \in \{0,1,2,4\}$
- Metrics: EE/elbow velocity-profile MSE · release timing · action jerk
- Pass: match $\gg$ zero on dynamic subset; NOT required: real 105/120
- Fail modes: quasi-static-only sweep; wrong sign; different ckpts; ignore grip $\neq$ arm latency

Fig. 5 — latency interface

Obs sync · DP chunk · discard stale · send-ahead (ex. arm 100 ms / grip 120 ms)

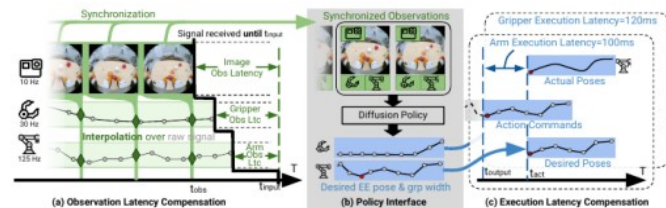


Fig. 5: UMI Policy Interface Design. (b) UMI policy takes in a sequence of synchronized observations (RGB image, relative EE pose, and gripper width) and outputs a sequence of desired relative EE pose and gripper width as action. (a) We synchronize different observation streams with physically measured latencies. (c) We send action commands ahead of time to compensate for robots' execution latency.

dynamic actions such as tossing (shown in Fig. 7). In addition, the joint visual-inertial optimization allows direct recovery of real metric scale, important for action precision and inter-gripper pose proprioception PD2.3: a critical ingredient to enable bimanual policy.

HD5. Continuous gripper control. In contrast to the binary open-close action used in prior works [41, 44, 54], we found commanding gripper width continuously significantly expands the range of tasks doable by parallel-jaw grippers. For example, the tossing task (Fig. 7) requires precise timing for releasing objects. Since objects have different widths, binary gripper actions will be unlikely to meet the precision requirement. On UMI gripper, finger width is continuously tracked via fiducial markers [16] (Fig. 2 left). Using series-elastic end effectors principle [42], UMI can implicitly record and control grasp forces by regulating the deformation of soft fingers through continuous gripper width control.

HD6. Kinematic-based data filtering. While the data collection process is robot-agnostic, we apply simple kinematic-based data filtering to select valid trajectories for different robot embodiments. Concretely, when the robot's base location and kinematics are known, the absolute end-effector pose recovered by SLAM allows kinematics and dynamics feasibility filtering on the demonstration data. Training on the filtered dataset ensures policies comply with embodiment-specific kinematic constraints.

Putting everything together. The UMI gripper weighs 780g, with an external dimension of $L310mm \times W175mm \times H210mm$ and finger stroke of 80mm. The 3D printed gripper has a BoM cost of \$73, while the GoPro camera and accessories total \$298. As shown in Fig. 2, we can equip any robot arms with a compatible gripper and camera setup.

B. Policy Interface Design

With the collected demonstration data, we can train a visuomotor policy that takes in a sequence of synchronized observations (RGB images, 6 degrees-of-freedom end-effector pose, and gripper width) and produces a sequence of actions (end-effector pose and gripper width) as shown in Fig. 5

(b). In this paper, we use Diffusion Policy [9] for all of our experiments, while other frameworks such as ACT [53] could potentially serve as a drop-in replacement.

An important goal of UMI's policy interface design is to ensure the interface is **agnostic to underlying robotic hardware platforms** such that the resulting policy, trained on one data source (i.e., hand-held gripper), could be directly deployed to different robot platforms. To do so, we aim to address the following two key challenges:

- **Hardware-specific latency.** The latency of various hardware (streaming camera, robot controller, industrial gripper) is highly variable across system deployments, ranging from single-digit to hundreds of milliseconds. In contrast, all information streams captured by UMI grippers have zero latency with respect to the image observation, thanks to GoPro's synchronized video, IMU measurements and the vision-based gripper width estimation.
- **Embodiment-specific proprioception.** Commonly used proprioception observations such as joint angles and EE pose are only well-defined with respect to a specific robot arm and robot base placement. In contrast, UMI needs to collect data across diverse environments and be generalizable to multiple robot embodiments.

In the following sections, we will describe three policy interface designs that address these challenges.

PD1. Inference-time latency matching. While UMI's policy interface assumes synchronized observation streams and immediate action execution, physical robot systems do not conform to this assumption. If not carefully handled, the timing mismatch between training and testing can cause large performance drops on dynamic manipulation tasks that require rapid movement and precise hand-eye coordination, demonstrated in Sec V-B. In this paper, we separately handle timing discrepancies on the observation and action sides:

PD1.1) Observation latency matching. On real robotic systems, different observation streams (RGB image, EE pose, gripper width) are captured by distributed micro-controllers, resulting in different observation latency.

100/120 ms are EXAMPLES not universal constants — measure per install (App. A)

Latency calibration recipe (App. A)

Always budget a calibration day — not optional for toss-class

1 Camera

Film rolling QR of system clock $\rightarrow l_{\text{cam}} = t_{\text{recv}} - t_{\text{display}} - l_{\text{display}}$

2 Proprio

HW stamp delta or $\approx \frac{1}{2}$ RTT when no hardware timestamp

3 Exec

Cross-correlate commanded vs measured sinusoid/teleop $\rightarrow l_{\text{action}} = l_{\text{e2e}} - l_{\text{obs}}$

4 Run

Sync streams to cam (highest latency); drop stale; command ahead by l_{action}

Motion deep dive — $T_a=6$ DP chunks

Relative not abs/delta · latency first-class for dynamic

$T_a = 6$

action horizon

10 Hz

cup/fold/wash/wild

20 Hz

tossing only

$I\text{-}T_o \in \{1,2\}$

obs image horizon

$P\text{-}T_o = 2$

proprio horizon

DDIM 50/16

train / infer steps

0.5×

quasi-static exec

1.0×

toss release vel

H-commit: effective commit $\approx T_a \cdot \Delta t$ near DP Push-T morals ($\sim 0.5\text{--}1$ s) for quasi-static; shorter / better
latency-matched for toss

Hypotheses (falsifiable)

Success = directional ranking, not paper parity

H-rel Relative traj → lower SE(3) err + higher proxy SR than abs; $\geq$ delta (paper 100/25/80)

H-lat-dyn Disable latency → sharp drop on dynamic toss-proxy (−30 pp moral); mild on grasp

H-lat-curve U-shape: under- and over-compensation (wrong sign) hurt velocity tracking

H-emb EE-relative + wrist FoV transfers across kinematics iff kin-filter at ingest

H-dp-umi Short DP smoke fits multimodal human modes better than MSE-MLP on same shard

H-bimanual Without inter-gripper relative proprio, hem-lift sync fails (70→30)

H-wild Multi-scene handheld mix beats single-scene @ matched N on scene-OOD cup

H-commit $T_a \cdot \Delta t$ near 0.5–1 s quasi-static; toss needs shorter/better latency match

RC0-RC4 smoke cards

Quiet mode = outlines only · do NOT sbatch from box

RC0

demo_ingest

CPU <30m
schema · rel Identity@t0
kin-filter report

RC1

relative_abl

1 GPU · 2-8 h
{rel,abs,delta}
#1 PRIMARY

RC2

latency_sweep

≤1-4 GPU-h
same ckpt toggle
#2 PRIMARY

RC3

policy_train

4-12 GPU-h
To=2 Tp≈16 Ta=6-8
DDIM 10-16 scout

RC4

embod_stub

CPU / <1-3 GPU-h
FK/IK A vs B
feasibility CSV

NOT required: paper 100%/87.5%/70% real-robot rates · 3 seeds · full 305+280+250+258 retrain · physical bins

GPU honesty — Table A1

Cite epochs/GPUs only · do NOT invent wall-clock hours

Task	Ta	Freq	Speed	Vision	Epochs	Batch	GPUs
Cup lab	Ta=6	10 Hz	0.5×	ViT-B/16	250 ep	bs 512	4×A10g
Toss	Ta=6	20 Hz	1.0×	ResNet-34	350 ep	bs 1024	4×A10g
Cloth	Ta=6	10 Hz	0.5×	ResNet-34	100 ep	bs 1024	4×A10g
Dish	Ta=6	10 Hz	0.5×	ViT-B/16	90 ep	bs 224	8×A10g
Cup wild	Ta=6	10 Hz	0.5×	ViT-L/14	50 ep	bs 512	8×A100

HONESTY RULES

Wall-clock hours NOT reported in paper — TENTATIVE if anyone quotes hours. Course smoke: 1 GPU, few hours, directional. Full wild ViT-L = 8×A100 class — Partial proxy only.

Wild OOD + failure modes

Fig. 9: $1400/30 \rightarrow 43/60 = 71.7\%$ · narrow ViT = 0%

71.7%

wild OOD 43/60

0%

narrow-only OOD

1400 / 30

demos / locations

12 p-h

3 demonstrators

SLAM texture

Textureless walls break VIO — map carefully; optional mapping fiducials

Sunlight

README known OOD for wild cup ckpt under direct sun

Mirrors raw

Without digital reflect+swap, mirrors HURT (85% < 90% none)

Abs calib bias

Absolute EE + bad SLAM-base calib → biased policy (25%)

Joint limits

FR2 fails 2/20 if base mount wrong — kin-filter post-hoc

Jaw ceiling

Parallel-jaw only — no dexterous hand transfer story

Empiricist — close

Takeaways + exam bite

1. RC1 relative + RC2 latency are PRIMARY smokes — directional ranking, not paper parity.
2. Reproduce: mamba umi → slam_pipeline → train.py → eval_real; wild zarr/ckpt public.
3. GPU honesty: Table A1 epochs/GPUs only; never invent wall-clock hours.
4. Tattoo: cup 100/90 · toss 87.5/57.5 · wild 71.7 vs 0 · rel 100 vs abs 25 · Ta=6 @10Hz.

EXAM BITE

UMI = in-the-wild WITHOUT in-the-wild robots; relative EE + latency + fisheye wrist; DP Ta=6; OOD cup
~72% after 1400/30.